

# Psychoacoustic Study of Drone Noise Perception

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## ABSTRACT

This study presents the psychoacoustics of a drone's propeller noise under multiple configurations. The measurements of acoustic pressure signals have been performed in an anechoic room by an array of microphones. Those recorded signals are then processed to compute the main 6 psychoacoustic metrics: Loudness, Fluctuation Strength, Annoyance, Roughness, Tonality and Sharpness. By comparing these results to the configuration changes, it is possible to deduct which parameter influences the psychoacoustic metrics the most. Results indicate that rotational speed is a dominant factor in perceived loudness, hence annoyance. The number of blades influences tonal characteristics, while beam position and diameter affect significantly the directivity of the noise generated. A closer and/or wider beam result in a higher annoyance. Those results show the importance of structure optimization to lower drone noise.

## Nomenclature

|           |                                                        |
|-----------|--------------------------------------------------------|
| $\theta$  | Azimuth angle (degrees)                                |
| $\varphi$ | Latitude angle (also defined as polar angle) (degrees) |
| SQAT      | Sound quality analysis toolbox                         |
| SQM       | Sound quality metric                                   |
| BPF       | Blade passing frequency (Hz)                           |
| rpm       | Rotations per minute                                   |
| SPL       | Sound pressure level (dB)                              |
| UAV       | Unmanned air vehicle                                   |

## **1. Introduction**

In 1917, the very first unmanned aerial Vehicle (UAV) was invented for war but never used. It is only from the late 20<sup>th</sup> century that drones have been slowly adopted in non-military domains, first used in scientific applications (weather monitoring, environmental research, wildlife monitoring...) and from the early 2000's to now in every applicable domain, like agriculture, cinematography, delivery services, infrastructure inspections, medical or even urban mobility.

The commercial drone market has been expanding over the past decade, from niche war technology to a low-cost tool across many industries. What started as a device for aim-training has transformed into a tool that companies use to boost efficiency, reduce costs, and improve safety.

As the use of drones became more popular, it is primordial to reduce the noise pollution generated by drones.

For civil applications, the European Union Aviation Safety Agency (EASA) and the Federal Aviation Administration (FAA) are introducing restrictions to minimize the noise in urban areas.

For military applications, reducing drone noise means increasing recognition capability with lowering the detection risk.

For delivery services or urban air mobility, reducing noise drone significance a key challenge to become a viable solution for our future.

Even if planes are louder than drones, the drone noise is usually reported as more irritating due to its high frequency. It also has an impact on the wildlife, birds and other ecosystems.

But how is noise pollution defined? How geometric parameters and configurations affect noise? Which psychoacoustics criteria best explain the perception of annoyance?

Drone noises are generated from multiple sources, in this study we will be focusing on the noise emitted from the interaction between the blade, air and a beam.

To describe the noise generated from those sources, it is necessary to use specific metrics. Most of the conventional sound metrics used in science are based on the sound pressure level, conventional metrics do not represent how the sound is perceived by a human. The employed metrics in this study

represent our feeling and sensation in reaction to a sound. Those metrics are calculated with weighting function and models created with series of tests and models.

Even if the evaluation of rotor noise is not a new topic ([George, 1978](#); [Brentner and Farassat, 1994](#) [7]) most studies are related to helicopters and as helicopters operate at high Mach numbers, UAVs generally operate at lower Reynolds and Mach numbers. Resulting in the non-possibility of applying the existing knowledge and expertise from helicopters to UAVs. In this study, we will process and analyze the noise generated by a propeller of a drone with a NACA profile blade in an anechoic room, using an array of microphones under different configurations.

Firstly, the main noise source and the five psychoacoustic criteria that were used during this study will be defined.

Secondly, the experimental set up that was used to measure the drone noise will be detailed, all the aspects will be defined to have a better understanding of the later results., providing a clear understanding of the parameters influencing the acoustic response.

Next, the development of programs and the use of computational tools for analysis will be presented.

Consequently, the results will be analyzed and discussed, pointing out trends and implications.

Finally, a conclusion will be made with perspectives of possible studies.

The objectives of this study are firstly to have a better understanding of the parameters that make a drone noisy and annoying to a human. Another goal is to apply one of the models of annoyance proposed by literature and analyze the results. Lastly, after completing those objectives, it is possible to propose further studies and points of improvement to reduce the drone noise.

## **2. Experimental Set up**

### 2.1. Anechoic chamber

Acoustic measurements were conducted in the anechoic chamber of ISAE-SUPAERO. Thanks to this chamber, the noise measured is considered as a free field without any echo and disturbance coming from airflow recirculation. This anechoic room is treated acoustically in the range 80 Hz to 16 000 Hz and has wedge-tip to wedge-tip dimensions of 5.02m long, 5.24m wide, and 5.34m high.

### 2.2. Propeller set up

The Faulhaber (Schönaich, Germany) 3274G024BP4 3692 electric brushless motor with low noise emission was used during this experiment to have a

better isolation of the propeller noise. The rotor was placed in the middle of the chamber at the top of a rotor test stand, as can be seen in **Figure 1**. The propeller can be changed easily to adapt the number of blades, the rotor used to conduct the measurement feature a NACA0012 blade cross section profile extruded in the radial direction with a constant chord of 25 mm, a constant pitch of 10°, and a diameter of 200 mm. Lastly, a circular shaped beam can be added under the propeller with a position that can be changed vertically to simulate the beam that connects the motor to the body of the drone, finally the rotational speed of the propeller can be adapted.

### 2.3. Acoustic measurements

A directional antenna equipped with thirteen 1/4-inch GRAS (Holte, Denmark) 40PH microphones is used to measure far-field noise at 1.62 meters from the rotor center, for latitude angles. Measurements are taken every 10° from -60° to 60°, with 0° corresponding to the rotor disk plane. In addition to the latitude angle, the array of microphones is set on a turntable that can rotate around the beam-propeller set up. With this configuration, the directivity will be studied with azimuthal angle of 10°. Every single microphone acquires acoustic data with a sampling of 51.2kHz during 16 seconds after initial transients.

The measurements have been conducted with these variables; the following array, in **Table 1**, represents a summary of every parameter that is studied:

| Numbers of blades | Rotational speed (rpm) | Beam diameter (mm) | Beam position (mm) |
|-------------------|------------------------|--------------------|--------------------|
| 2                 | 6000                   | 2                  | 2                  |
| 2                 | 7000                   | 2                  | 2                  |
| 2                 | 8000                   | 2                  | 2                  |
| 2                 | 8000                   | 2                  | 3                  |
| 2                 | 8000                   | 2                  | 4                  |
| 2                 | 8000                   | 1                  | 2                  |
| 2                 | 8000                   | 1,5                | 2                  |
| 2                 | 8000                   | None               | None               |
| 3                 | 8000                   | None               | None               |
| 4                 | 8000                   | None               | None               |

**Table 1: Summary of the different configuration used**

## 3. Methodology

### 3.1. Sound Quality Analysis Toolbox

To process the given temporal data into a measurement that will indicate the quality/annoyance of the noise, the toolbox Sound Quality Analysis toolbox (SQAT [4]) will be used. The SQAT permits to calculate various sound quality metrics :

**Loudness:** The loudness, in sone, describes the perceived magnitude of a sound in a linear scale based on human auditory perception.

**Fluctuation strength:** Expressed in vacil, this metric quantifies the perceived strength of slow temporal variations in a noise with a frequency up to 20 Hz.

**Roughness:** This metric, expressed in asper, defines the sensation caused by fast amplitude modulations of a noise with a modulation frequency in the range [15;300] Hz. Noises with higher roughness, such as buzzing or growling noises, are perceived as more aggressive.

**Tonality:** On a scale from 0 to 1, describes the prominence of tonal components in a noise depending on various factors such as bandwidth, center frequency, level above threshold, and the number of tonal components. The Aures tonality model is commonly used for its calculation.

**Sharpness:** Expressed in acum, the sharpness quantifies the high frequency content of a noise.

The metrics of the SQAT toolbox are using [the following models \[5\]](#) :

| Metric               | Model                   |
|----------------------|-------------------------|
| Loudness             | Loudness ISO 532-1:2017 |
| Sharpness            | DIN 45692:2009          |
| Roughness            | Daniel & Weber          |
| Fluctuation Strength | Osses et al.            |
| Tonality             | Aures                   |

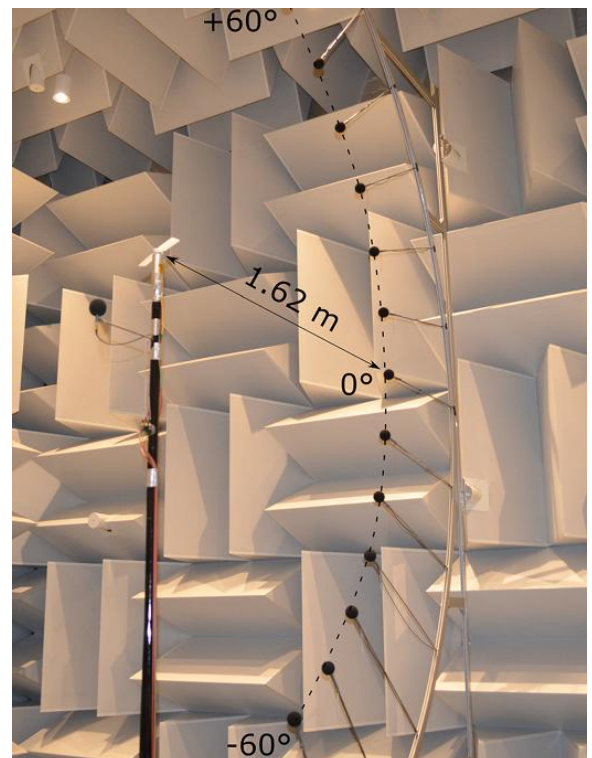
**Table 2: Model used to compute the metrics**

Furthermore, as recommended in literature (Zwicker and Fastl, 2013 [5]), the 5<sup>th</sup> percentile of each metrics will be used to analyze and compare data. The 5<sup>th</sup> percentile of a set of values is defined as the value below which 95% of the data points fall when the values are arranged in ascending order, meaning that 5% of the values are greater than this threshold. In addition to this percentile, the first 0.5 seconds of each metrics are ignored in order to avoid transient effect of the digital filters implemented in the SQAT algorithms.

### 3.2. Developed programs

After a recording session is done, the data is later saved to a temporal file, containing all thirteen latitude angles for each azimuthal angle. As this represents a large amount of data, it is needed to develop a program to concatenate all the available data into one single MATLAB structure, making the future analysis easier and introducing less mistakes. The concatenation program uses the SQAT toolbox to compute the 5<sup>th</sup> percentile of every metrics into a single structure.

To analyze the results, it was decided to use three different methods of visualization. The first one is a simple cartesian plot, allowing to understand the trends and the correlation between the varying parameter and the Sound Quality Metrics (SQM). The second one is a polar plot, helping to visualize the evolution of the metric on the latitude angle. The last program developed is a 3D visualization of each metrics around the propeller. With this last method, it is possible to have a full analysis of the metrics interacting with the environment.



**Figure 1 – Photograph of the experimental setup in the ISAE-SUPAERO anechoic chamber**

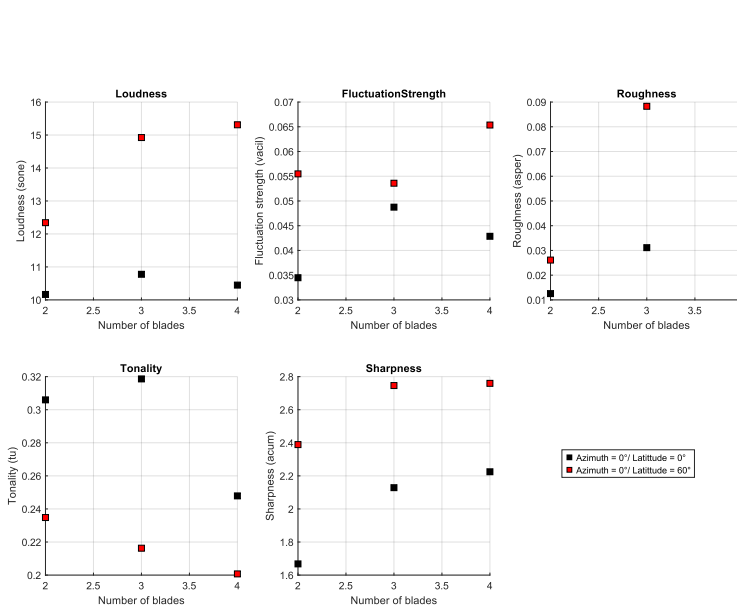


Figure 1 - Evolution of the metrics with the evolution of the number of blades

#### 4. Results Analysis

The evolution of the sound quality metrics will be analyzed with the variation of one parameter. This is done by using two different visual representations: a polar representation, allowing to display every elevation angle of the microphones, and a cartesian graph showing the evolution of each metrics with the variation of one parameter.

##### 4.1. Number of blades

The first parameter to be studied is the number of blades on the propeller. In this first configuration, the speed is kept constant at 8000 rpm, the blade profile used is a NACA0012 and no beam is placed under the propeller. This configuration ensures that the only interaction is between the air and the blades of the propeller.

Regarding the first metric: The loudness, it is possible to see in both **Figure 2** and **Figure 3** that the values are stable around 10.5 sone for an elevation angle of 0° and are the greatest with 13 to 15 sones for an angle of ±60°. These results suggest that the primary noise source is directed perpendicular to the propeller's axis as the only source of noise is emitted from the interaction between the air and the blade of the propeller. In the **Figure 2**, the values of the measurements with three and four blades show a similitude in the value range. The results show that increasing the number of blades beyond three does not have an impact on the loudness perceived. In conclusion, a propeller with two blades will be perceived less loud than with a three or four blade propeller and that the loudness will be greatly decreased when the listener is positioned along the same axis as the propeller.

In this configuration, the fluctuation strength and the roughness can be considered as equal to zero no matter what emission angle or no matter what number of blades. This suggest that the fluctuation strength is a parameter that is not linked to the interaction of the surrounding air and the propeller. It is suggested, in the future, to study the interaction of the blades, beam and air to determine if the number of blades have an impact on this metric with an obstacle.

The next metric to be studied is the tonality. As shown in **Figure 2**, the data measured with a latitude angle of 60° shows a slight decrease in tonality as the number of blades grows, from 0.24 aures to 0.2. This trend is attributed to the blade passage frequency, this frequency increases as more blades are introduced, and the energy associated with each harmonic is spread out, making the noise sound less tonal. This is confirmed by [Romain Gojon et al.](#)

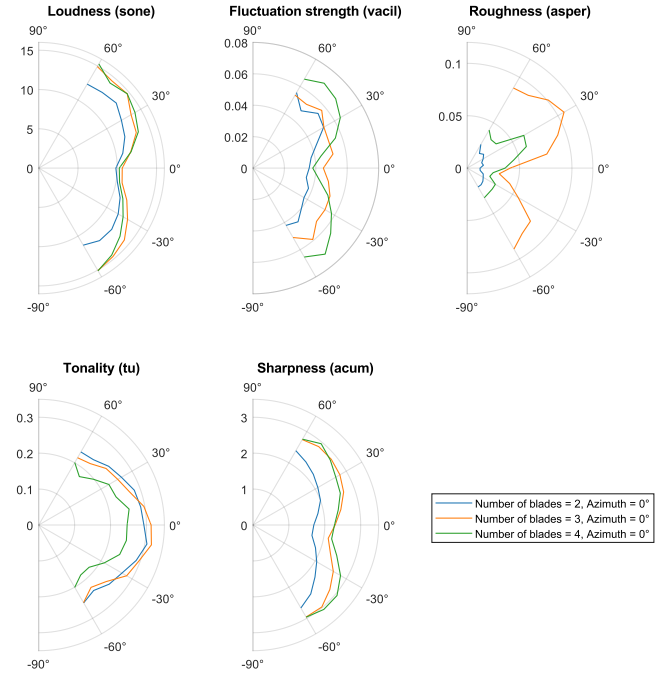


Figure 3 - Polar plots of the metrics with the evolution of the number of blades

Additionally, thanks to the **Figure 3**, it is important to note that the values for a two and three-rotor blade are relatively similar whereas a 4-rotor blade reduces the tonality of the noise.

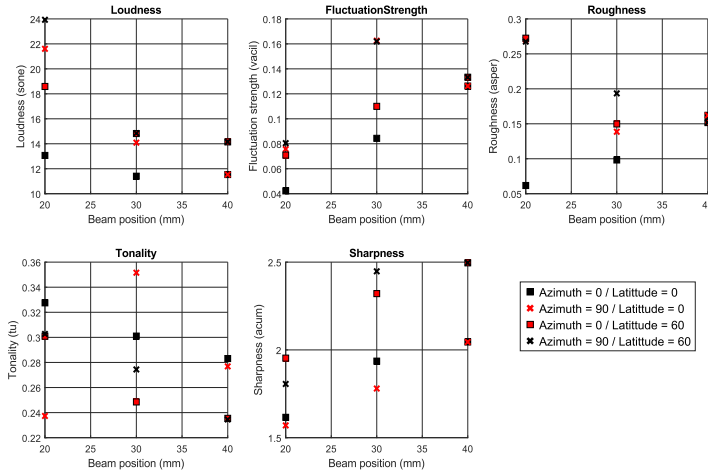
Finally, the sharpness is examined. Results on the **Figure 2**, indicate a growing trend correlated with the number of blades. The trend seems to grow steeply between two and three blades, beyond three blades, the value stabilizes.

In summary, the analysis of the propeller blade number reveals several trends. The loudness increases with the number of blades but stabilizes after three blades, the maximum value is always recorded at emission angle of ±60°. Fluctuation strength and roughness show low values in this configuration, suggesting that none of the varying parameters are influencing those metrics. Tonality decreases with an increasing number of blades due to the redistribution energy across the frequency spectrum. Finally, sharpness increases as more blades are introduced, particularly between two and three blades.

##### 4.2. Beam position

The impact of the beam position under the propeller is going to be analyzed, during this section. The only varying parameter is the beam position relative to the propeller axis, with measurements taken at 20 mm, 30 mm, and 40 mm. The different azimuth angles are also displayed in the graphs. The **Figure 4** presents the sound quality metrics previously defined in cartesian, and the **Figure 5** represent the polar graph.

First of all, the loudness shows on the **Figure 4** an overall decreasing trend with a pronounced decrease between the two positions 20mm and 30 mm after which loudness stabilizes. It is also important to note that the highest



**Figure 4 - Evolution of the metrics with the evolution of the beam position**

loudness values are observed when the azimuthal angle is 90°, which can be explained by the fact that the beam does not obstruct the noise propagating in this direction.

This means that the beam interacts, significantly, with the noise distribution when positioned closer to the propeller, leading to higher loudness values at lower beam positions. The differences in loudness across azimuth angles highlight the directional dependency of the noise propagation.

Next metric, the fluctuation strength shows different behaviors depending on the azimuth angle. For the dataset with an azimuth angle equal to 0°, the trend seems to be a linear growth as the distance of the beam increases. A maximum is reached for the greatest distance. However, for an azimuth angle of 90°, a peak can be found for a position of 30mm, suggesting a maximum interaction between the propeller and the beam for this metric. This nonlinear growth could be explained by the constructive interference effect or modification of turbulence patterns around the beam.

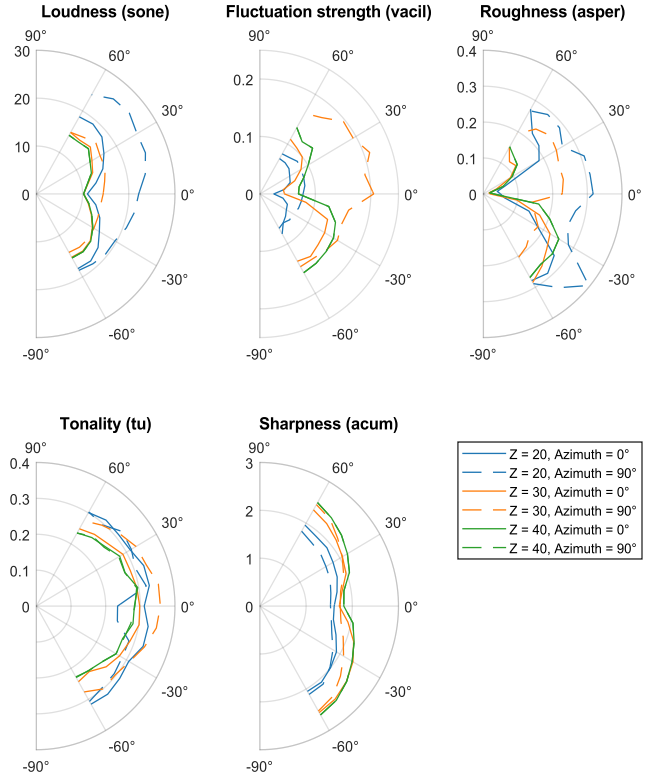
In summary, these results indicate that fluctuation strength is highly sensitive to spatial variations and to the position of the beam in the sound field.

For the roughness, there is no obvious trend in the results. The roughness fluctuates with the beam position, suggesting that the spatial distribution is affected. This is potentially due to aerodynamic phenomena such as a blade vortex interaction.

As the roughness is associated to fast modulations in the noise signal, it is very probable that the interactions between the blade and the beam introduce turbulence modulations.

Regarding the tonality, it is associated to its fundamental frequency. While the data show a decreasing trend of the tonality with the growth of the beam position, the data for an azimuth angle of 90° and aligned with the axis of the blade presents a maximum for a position of 30 mm under the propeller. The presence of high tonality values at a specific position suggests that certain beam positions align with dominant tonal sources, such as the blade passage harmonics.

Since tonality is associated with the presence of prominent frequency components, the changes observed in the data suggest that the beam affects the propagation or generation of dominant tonal frequencies. The peak at 30 mm suggests that at this position, there is a stronger reinforcement of tonal components, possibly due to the reflection or diffraction of specific frequency components.



**Figure 5 - Polar plots of the metrics with the evolution of the beam position**

Finally, the sharpness indicates that high frequencies of the noise are growing with the beam distances from the propeller axis. Specific configurations show a rapid growth such as the configuration where both azimuthal angle and elevation angle are equal to 90°.

This behavior indicates that high-frequency components are more pronounced when the beam is farther from the propeller. The increase in sharpness with beam distance suggests that the beam acts as a modulating element, affecting the spectral balance of the noise. The stronger presence of high-frequency components at larger distances may be attributed to the directivity patterns of the propeller noise, where high-frequency noise propagates more efficiently at oblique angles away from the immediate vicinity of the propeller.

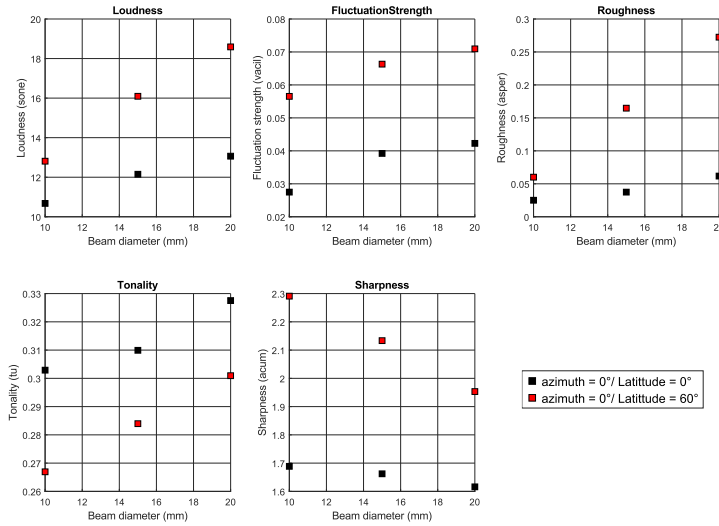
In summary, the results reveal a strong dependence of psychoacoustic metrics on beam position, with notable differences between azimuth angles. Loudness and tonality exhibit a decreasing trend with distance, while fluctuation strength and sharpness show distinct peaks for specific configurations. Roughness, on the other hand, fluctuates without a clear trend, likely due to aerodynamic interactions. These findings provide valuable insights into how beam placement affects perceived noise characteristics, offering potential implications for noise mitigation strategies in drone applications. The variation in psychoacoustic responses suggests that careful positioning of structural elements such as beams can influence the perceived noise levels and characteristics.

#### 4.3. Beam diameter

In this section the studied parameter is the influence of the beam diameter on the propeller. In this configuration, the speed is once again kept constant at 8000 rpm, the blade profile used is a NACA0012 and a beam is placed 20 mm under the propeller.

The graph in the **Figure 6** indicates a linear correlation between beam diameter and loudness. This suggests that a larger beam produce an interaction that is perceived louder. The trend is attenuated at elevation angles closer to the axis of the beam. An increase of 10 mm of the beam at a latitude angle of 60° results as an increase of almost 6 sone.

The fluctuation strength clearly follows the same trend as the loudness, nevertheless, the values remain low with a maximum value of 0.07 vacil for the largest beam at a latitude angle of 60°.



**Figure 6 - Evolution of the metrics with the evolution of the beam diameter**

For elevation angle of 0°, the roughness increases gently but stays under 0.05 asper. These phenomena and values are rapidly amplified for higher absolute values as shows the graph in the **Figure 7**. The linear trend grows to a value of 0.27 asper for latitude value of  $\pm 60^\circ$ .

Similarly to the other metrics, tonality values increase as the beam diameter increases. As a shown in the study of [Romain Gojon et al \[2\]](#), with a larger beam comes with presence of higher amplitude harmony frequency. As the Aures tonality model does not rely only on the amplitude of the fundamental frequency or BPF, it also evaluates peak prominence, narrow peak and impact on the loudness. This results in a higher tonality for larger beams.

In contrary, the sharpness shows an inverse relationship with beam diameter and the sharpness perceived. This trend is verified by the results found by the article of PARISOT on the spectral frequency graph. High frequency contents are amplified with a smaller beam, particularly seen at higher absolute latitude angles as suggest the **Figure 7**.

The study of the beam diameter has highlighted the influence on psychoacoustic metrics. For instance, the loudness increases with beam diameter, indicating stronger interactions, especially at larger latitude angles. Fluctuation strength follows the same pattern but remains low. Roughness increases slightly at lower angles and more significantly at larger latitude angles. Tonality also rises with beam diameter due to stronger harmonic components but has shown greater values on the axis of the beam.

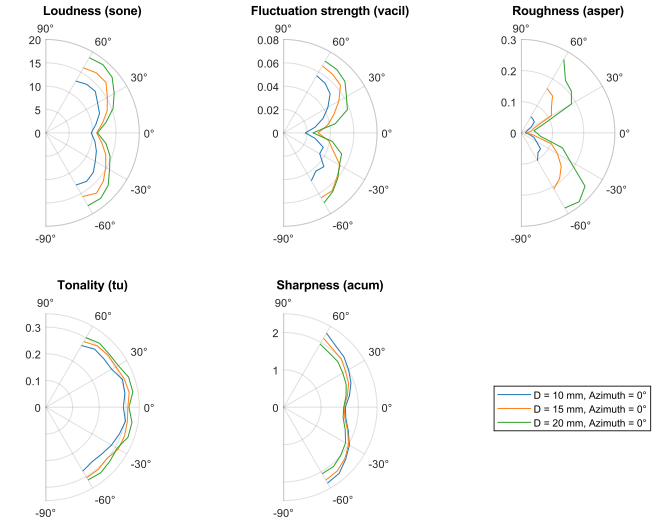
In contrast, sharpness decreases with beam diameter, as smaller beams amplify high-frequency content, particularly at extreme latitude angles. This aligns with spectral analyses showing enhanced high-frequency contributions for narrower beams.

Overall, a larger beam increases loudness, roughness, and tonality, while a smaller beam enhances sharpness. These findings show the importance of the beam diameter in noise perception.

#### 4.4. Rotational speed

In this configuration, the propeller has two blades, the blade profile is NACA0012, and a beam with a 20 mm diameter is placed 20 mm under the propeller. The study aims to understand how perceived noise characteristics evolve with different speeds, at 6000, 7000, and 8000 rpm.

Loudness is directly influenced by the rotational speed, as a higher speed generates increased aerodynamic forces and, consequently, greater sound pressure levels. Examining **Figure 8**, it is evident that loudness increases as the rotational speed increases linearly from 6000 to 8000 rpm. The measurements indicate a significant rise in loudness, with greater values appearing at azimuthal angles close to  $\pm 60^\circ$ . The trends validate our previous statement, greater rotational speed leads to a higher interaction between the propeller, the beam and the surrounding air, producing louder



**Figure 7 - Polar plots of the metrics with the evolution of the beam diameter**

acoustic emissions. The polar representation in **Figure 9** provides clear evidence that the loudest noise is perceived perpendicularly to the propeller axis. Therefore, the loudness is dependent to rotational speed and emission angle.

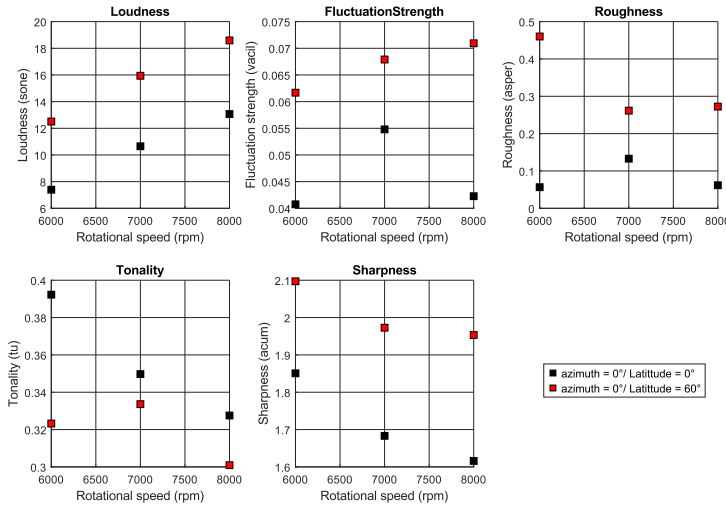
**Figure 8** and **Figure 9** reveal that fluctuation strength remains relatively low. However, a growing tendency is identified for an azimuthal angle of  $60^\circ$ , this suggests that at a maximal rotational speed, the periodic interaction between the blades and the beam positioned under the propeller becomes more pronounced, putting into evidence low-frequency modulations into the noise spectrum. This effect is attenuated at closer azimuthal angles where the beam is aligned with the main noise direction.

Roughness, in this configuration, appears to follow a decreasing trend with rotational speed. This effect is particularly notable at azimuthal angles where the blade's axis is not aligned. Moreover, **Figure 8** shows that the roughness is higher when perceived in negative emission angles. As the rpm are increased, the frequency components are shifted to higher values, knowing that the roughness is associated to fluctuations in the range [15;300] Hz, hence the lower roughness values.

The tonality analysis from **Figure 8** shows a decreasing trend as the rotational speed increases. This behavior can be attributed to the same phenomena described in the section where the number of blades is analyzed. The blade passage frequency is increased, which distributes the spectral energy over a larger range, lowering the prominence of tonal components. The trend is much clearer for an angle of emission equal to 0. In this direction, the tonal noise is not impacted by the beam. Nevertheless, the presence of the beam under the propeller affects the tonal components, decreasing the perceived tonality at higher speeds.

The effect of rotational speed on sharpness is evident in **Figure 8**, where a clear downtrend is observed. Sharpness is associated with the presence of high-frequency noise components, and as just explained for the last metric, intensifies with increasing speed. Resulting in a substantial increase in sharpness, particularly in emission directions perpendicular to the propeller axis. This observation aligns with the expectation that higher speeds generate more high-frequency content.

The study of rotational speed variations reveals several key trends in multiple psychoacoustic metrics. Loudness exhibits an expected strong dependency on speed. This trend is clearer especially in emission directions perpendicular to the propeller axis, where the values are the highest. Fluctuation strength values are low but show an increasing trend at specific emission angles, suggesting that with high speeds, the interaction between the blades and the beam becomes more pronounced. Roughness, in contrast, follows a decreasing tendency, at azimuthal angles where the blade's axis is not aligned. This phenomenon is due to the shift of dominant frequency components beyond the range associated with roughness



**Figure 8 - Evolution of the metrics with the evolution of the rotational speed**

perception. Similarly, tonality decreases as rotational speed increases, the results is caused by the spectral redistribution into a larger spectrum caused by higher blade passage frequencies, it is also noticed that the beam contributes to the reduction of tonal noise. Finally, sharpness validates the spectral redistribution with a decreasing trend.

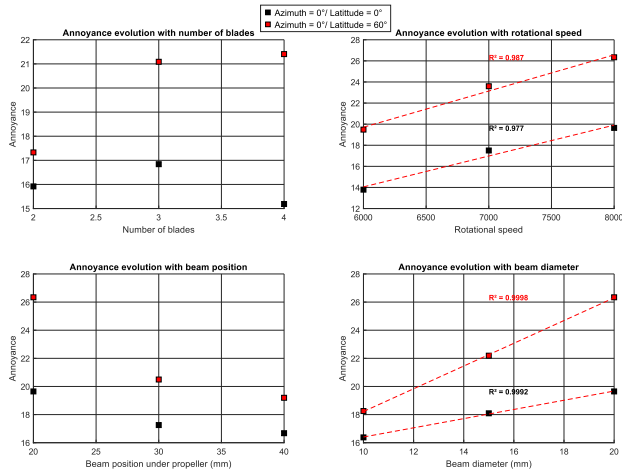
#### 4.5. Annoyance

The computation of the annoyance is also done thanks to the SQAT toolbox. The five calculated metrics (Loudness, Fluctuation strength, Tonality, Sharpness and Roughness) are used to compute a single parameter defined as the annoyance (PA). The annoyance is then computed using the improved psychoacoustic annoyance: [Di's Model \[6\]](#).

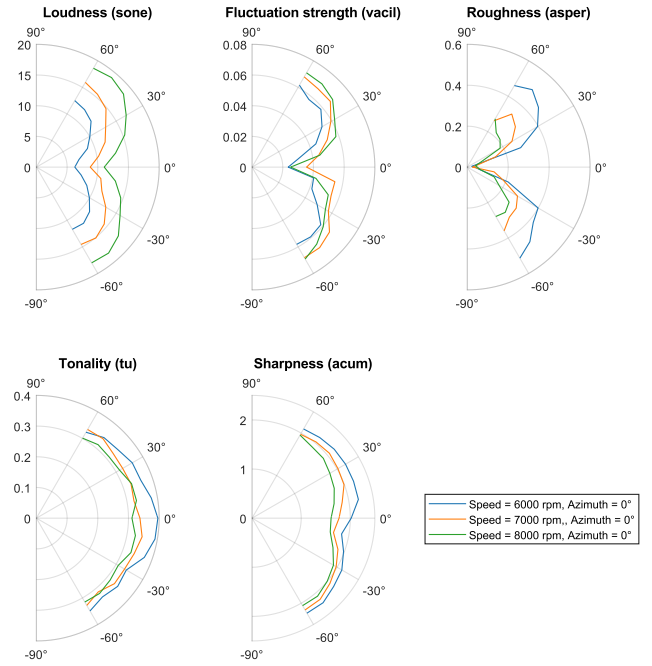
It introduces modifications to better estimate the annoyance caused by sounds with prominent tonal components. It is a model proposed by G. Q. Di and their colleagues to improve previous models (Zwicker's). This model considers the five metrics computed until now.

The annoyance is computed for all four varying parameter and displayed in the **Figure 10** and **Figure 11**.

The first graph in the **Figure 10** and **Figure 11**. Represents the annoyance with the evolution of the number of blades. It is possible to see that the number of blades has a direct impact on the perceived annoyance on an



**Figure 10 - Evolution of the annoyance with the evolution of all the parameters**

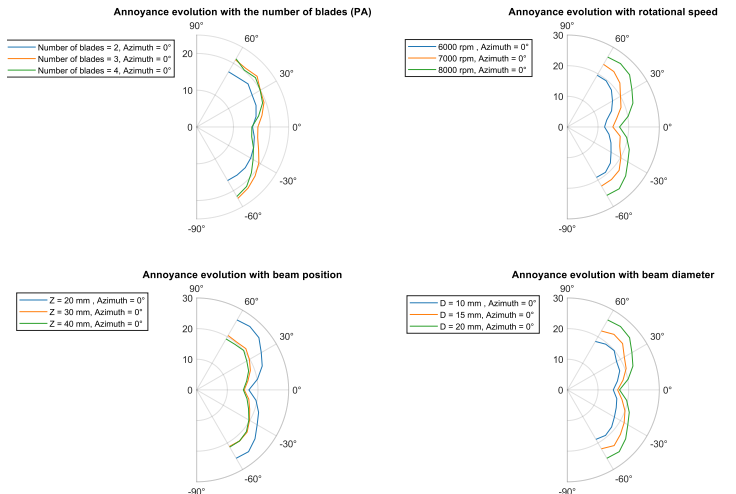


**Figure 9 – Polar plots of the metrics with the evolution of the rotational speed**

elevation angle superior to 10°. The trend shows a greater annoyance for a 3-blades propeller than with a two-blade propeller. For an elevation angle of 60°, the annoyance rises by nearly 8 PA from two to three blades and then, seems to stabilize, suggesting that, beyond three blades, the annoyance will be perceived equally. This augmentation is not to be ignored as it represents an augmentation of roughly 45%.

The following graphs represent the annoyance evolution with the configuration of a 20 mm diameter beam placed 20 mm under the two-blade propeller axis, and the rotational speed varying. Both graphs in **Figure 10** and **Figure 11** clearly put in evidence the linear relations between the rotational speed and the annoyance perceived. On this configuration, the annoyance grows from a 6000 rpm speed to a 8000 rpm speed with an augmentation of 6 PA no matter what the angle of emission is. The of annoyance for an azimuthal angle of 60° are the maximum values and represent a maximal augmentation of 30%.

The graphs introducing the evolution of the annoyance with a varying position of the beam under the propeller show a nonlinear tendency to decrease as the distance between the propeller and the beam grows. As it is possible to see on **Figure 10**, this decrease of the annoyance is the greatest for a latitude angle of  $\pm 60^\circ$ , resulting in the loss of nearly 30% of its original value.



**Figure 11 – Polar plots of the annoyance with the evolution of all the parameters**

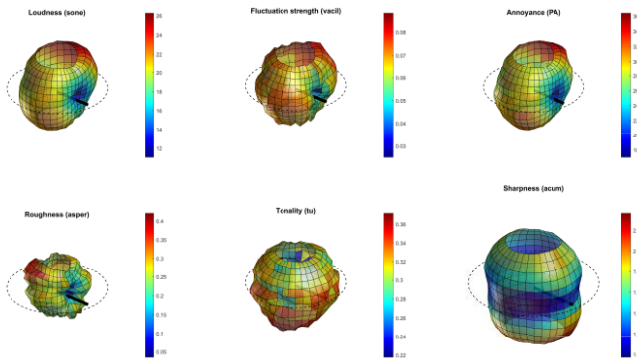
Finally, the last graph shows a linear correlation between the annoyance and the beam diameter. The annoyance grows with the beam diameter, but it is interesting to note that the trend does not have the same growth factor on the latitude angles. The trend seems to grow quicker for higher elevation angles. On the contrary, on the level of the axis of the propeller, the trend growth factor is lower.

In conclusion regarding the annoyance study, it has been shown that the annoyance increases when transitioning from a two-blade propeller to a three-blade propeller but beyond this, the perceived annoyance is equal. Suggesting that additional blades do not contribute to a bigger discomfort. In the same way, annoyance follows a linear correlation with the rotational speed, with a constant increase of 6 PA no matter the emission angle between 6000 rpm and 8000 rpm.

Although the position of the beam does not follow a linear trend, it plays a crucial role in reducing annoyance. As the distance between the beam and the propeller increases, the annoyance decreases. This reduction reaches 30% from its initial value at higher latitude angles, around  $\pm 60^\circ$ .

Lastly, the diameter of the beam has a direct influence on the annoyance with a linear growth; larger diameter causes bigger discomfort. The growth of the value depends on the elevation angles, with higher elevation angles, a faster growth rate of the trend is observed.

#### 4.6. Directivity



**Figure 12 - 3D representations of the metrics directivity**

As it has been shown in the previous parameters, the values are in average higher for an azimuthal angle different to 0. Therefore, all the sound quality metrics, including the annoyance, are displayed on 3D graphs. The measurements have been done with a beam placed 20 mm under the two-blade propeller and with a diameter of 20mm, the rotational speed has been kept at 8000 RPM. In this configuration, the data was every  $10^\circ$  on a azimuthal range of  $360^\circ$ .

On the **Figure 12**, the loudness distribution confirms the previous results. The loudness measured in the axis of the beam is minimal as it acts as an obstacle to propagate the noise. The highest values are determined at around 25 sone in the upper region. This is not an expected result and would need further investigation, such as blade geometry or airflow study.

The directivity of the fluctuation strength is coherent with the low results value. The highest values, reaching 0.08 vacil, appear in upper and lateral areas, while lower values (0.03 vacil) are concentrated in the axis of the beam.

Roughness is more localized, with peak values reaching 0.4 asper in specific upper areas. These localized maximums suggest a higher mid-frequency modulations level, caused by the interaction between the blade and the air. Tonality shows a more uniform distribution, with peak values around 0.36 tu. This suggests that tonal components are well distributed and do not seem to be interfered by the beam in any direction. The shape of the directivity is coherent with the work of the article [Romain Gojon et Al \[1\]](#), the tonality is associated closely to the blade passage frequency.

The directivity of the sharpness presents a double hump pattern with a minimum of 1.5 acum on the level of the blade, and peak values above 2.1 acum in the lower region. This suggests that high-frequency noise propagation is influenced by the vertical structure of the propeller. The elevated sharpness levels could indicate strong blade tip noise or higher-order harmonics, which are known to contribute to annoyance. Once again,

those results can be verified thanks to the work of [Romain Gojon et al \[2\]](#), as the directivity shape seem to be similar to the BBN.

The annoyance directivity has the same pattern as the loudness, with a minimum annoyance in the axis of the beam and a maximum on the upper section.

The spatial variations in these metrics show the non-uniform directivity of the propeller noise. The unexpected higher loudness above the drone suggests an influence of aerodynamic effects, reflections, or directional frequency propagation. Further analysis of frequency content, directivity mapping, and beam positioning must be conducted to have a better understanding of this phenomenon.

#### 4.7. Discussion

With these results, describing which configuration is the least annoying and/or which configuration minimizes a specific metric is clear. But this study does not show the exact reality. For instance, as shown by PARISOT, the produced thrust is bigger with the number of blades, meaning that to produce the same thrust as a two-blades propeller, a three or 4-blades propeller needs less rotational speed to reach the same level of thrust. As seen during this study, the rotational speed has one of the biggest impacts on the annoyance, this could mean that for an iso-thrust, the level of annoyance could be lower for a 4-blades propeller than for a two blades propeller.

The configuration of the experimental set up is perfect for a study like this one but presents a couple limitations. Firstly, the anechoic room enabled to represent a free field environment, hence eliminating any reflection or background noise. Future studies should consider performing outdoor measurements to assess the impact of environmental factors. Another limitation is the use of a single propeller design during the entire study. For future studies, it would be interesting to consider analyzing the impact of the geometry and the blade material on psychoacoustic metrics.

One of the biggest limitations of this study is the amount of data used: Each configuration was tested with only three different values of the varying parameter. While three values are bare minimum to see an initial trend, it is not enough sufficient to validate those findings.

Moreover, the time taken to compute the whole range of the latitude microphones for one configuration and one azimuthal angle is approximately 1.5 hours. It was not possible to process every azimuthal angle for every configuration, but it would have been interesting to see the evolution of the different metrics directivity as the set-up parameter varies. Furthermore, the varying parameters were studied independently, a suggestion is made to study every varying parameter in every configuration. For instance, study the evolution of the SQM with a beam diameter of 10; 15 and 20 mm at every distance from the blade. This would represent the next step identify optimal configurations that minimize annoyance across different condition.

Another key limitation of this study is the consideration of only a part of the drone, as drones are usually composed of four propellers linked together, the interactions between the beam and the complex airflow are not considered.

Finally, this entire study has been conducted with a static propeller, whereas drone applications in real world are in constant motion. The influence of Doppler effects and dynamic flight conditions should be explored in future studies.

To address those limitations, for further studies, the following research directions are proposed. Conducting similar studies in urban or rural environments with multi-propeller interactions to provide more applicable results for regulations and improvements.

Secondly, expanding the amount of data by increasing the number of tested values for each parameter, allowing statistic and trend validations.

The next suggestion is the studying of a full drone setup with multiple propellers to allow the comprehension of how beam placement and airflow interactions influence overall noise emissions. Finally, dynamic flight condition analysis should be investigated, focusing on how Doppler effects have an impact on the human noise perception.

By addressing these research directions, future studies can provide a more complete understanding of drone noise perception, completing this study.

## 5. Conclusion and Perspectives

This study helped to get a better understanding of the psychoacoustic perception of drone noise by investigating various parameters, including rotational speed, number of blades, beam position, and beam diameter. The findings highlight that rotational speed is a dominant factor influencing loudness resulting in impacting the annoyance, with higher speeds leading to increased perceived discomfort. The number of blades also plays a significant role, where an increase beyond three blades results in diminishing returns in terms of noise perception. Furthermore, beam positioning and diameter substantially affect psychoacoustic metrics, revealing the importance of structural elements in noise generation and propagation.

Several strategies for noise mitigation have been identified, including optimizing the number of blades, adjusting rotational speeds to balance thrust efficiency and noise reduction, and modifying beam placement to reduce unwanted interactions. The application of an annoyance model successfully quantified perceived discomfort, offering a valuable tool for future evaluations of drone noise perception. Additionally, the experimental results align with existing literature, confirming established trends while also introducing new insights specific to drone acoustics.

Despite the controlled conditions of this study, further research is required to address real-world scenarios. Future investigations should focus on outdoor noise propagation, the influence of multiple propellers in a full drone setup, and dynamic flight conditions to capture the effects of movement and Doppler shifts. By incorporating these aspects, researchers can develop a more holistic understanding of drone noise perception and contribute to designing quieter, more acceptable drones for various applications, including urban mobility, delivery services, and environmental monitoring. In addition, it would be valuable to develop a perception-model through extensive listening tests, allowing for a more accurate prediction of annoyance and psychoacoustic responses to drone noise on a large amount of people to represent the population.

Ultimately, this study provides a solid foundation for future work in drone noise perception and mitigation. By refining measurement techniques, expanding test conditions, and incorporating human perception data, future research can lead to significant advancements in reducing drone noise pollution, ensuring better integration of drones into modern society while minimizing their acoustic impact.

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